Statistical Terms Glossary

COVID-19 Survival Prediction Project

A comprehensive reference for all statistical, machine-learning, and clinical terms used in the modelling pipeline.

Dataset	Synthea 10k Synthetic COVID-19 — 9,106 patients
Features	23 clinical, demographic, and lab features per patient
Target variable	60-day survival (Survived = 1 / Died = 0) | 96.1% : 3.9%
Best model	XGBoost — AUC-ROC: 0.9980 | PR-AUC: 0.9999
Sections	16 thematic sections covering the full ML pipeline

Table of Contents

Full metric comparison: LR vs. RF vs. XGBoost

1. Dataset & Sampling Terms

1.1 Train-Test Split

Item	Detail
Definition	Divides the dataset into a training set (used to learn patterns) and a test set (used to evaluate on unseen data). Standard practice to detect overfitting.
Parameters	test_size=0.20 → 80% train, 20% test; stratify=y → preserves class proportions
Code	pipeline.py — step5_preprocess() › train_test_split(X, y, test_size=0.20, stratify=y, ...)
Output	Train: 7,284 patients — Test: 1,822 patients (run_log.txt line 65)

1.2 Stratified Split

Item	Detail
Definition	A variant of train-test split that ensures each split contains the same proportion of each class as the full dataset. Critical for imbalanced data.
Why it matters	Without stratification on a 24.7:1 imbalanced dataset the 20% test set could accidentally contain very few 'Died' patients, making evaluation meaningless.
Output	Test set: 71 Died + 1,751 Survived — proportional to the original 3.9% death rate.

1.3 Random State / Random Seed

Item	Detail
Definition	A fixed integer seed passed to all random processes so results are exactly reproducible across runs.
Code	pipeline.py line 40: RANDOM_STATE = 42 — used in train_test_split, SMOTE, all three models.
Output	All results in output.txt are identical on every re-run as long as data does not change.

2. Feature Engineering Terms

2.1 Binary Feature (One-hot Encoded Flag)

Item	Detail
Definition	A feature that takes only the values 0 or 1, encoding absence or presence of a condition.
Code	pipeline.py feat_comorbidities() e.g. pre_covid['has_diabetes'] = desc.str.contains('diabetes').astype(int)
Features	has_hypertension · has_diabetes · has_obesity · has_asthma · has_COPD · has_heart_disease · was_hospitalized · is_icu · had_care_plan · low_spo2 · high_crp · high_d_dimer · low_lymphocytes

2.2 Derived Binary Threshold Flags

Item	Detail
Definition	Binary features derived from a continuous lab value by applying a clinical cut-point. Converts raw measurement into a clinically meaningful signal.
Thresholds used	low_spo2 = (SpO2 < 94%) · high_crp = (CRP > 10 mg/L) · high_d_dimer = (D-Dimer > 500 ng/mL) · low_lymphocytes = (Lymphocytes < 1.0 ×10³/µL)
Clinical basis	SpO2 < 94% = hypoxia (WHO) · CRP > 10 = active inflammation · D-Dimer > 500 = thrombotic risk · Lymphocytes < 1.0 = lymphopenia

2.3 Comorbidity Count

Item	Detail
Definition	A count feature summing all active binary comorbidity flags per patient. Captures additive disease burden in a single numerical feature.
Code	pipeline.py line 138: agg['comorbidity_count'] = agg[cols].sum(axis=1)
Stats	Mean = 1.04 | Max = 5 comorbidities per patient

2.4 Median Imputation

Item	Detail
Definition	Filling missing values with the median of observed values for that lab. The median is preferred over the mean because it is robust to outliers common in lab data.
Code	pipeline.py lines 175-178: median_val = lab_df[lab].median() | lab_df[lab] = lab_df[lab].fillna(median_val)
Values	d_dimer: 7,353 missing → median 0.300 · ferritin: 7,353 → 423.300 · crp: 7,353 → 10.200 · wbc: 6,206 → 3.800
Zero imputation	il6, spo2, lymphocytes had no data at all → imputed with 0 (WARNING in run_log.txt)

2.5 Feature Matrix (Design Matrix X)

Item	Detail
Definition	The rectangular table of shape (patients × features) fed into ML models. Each row is one patient; each column is one feature.
Shape	(9,106 patients × 23 features) (run_log.txt line 63)
Feature list	age · gender · has_hypertension · has_diabetes · has_obesity · has_asthma · has_COPD · has_heart_disease · comorbidity_count · d_dimer · il6 · ferritin · crp · spo2 · wbc · lymphocytes · low_spo2 · high_crp · high_d_dimer · low_lymphocytes · was_hospitalized · is_icu · had_care_plan

3. Preprocessing & Scaling Terms

3.1 Standard Scaler (Z-score Normalisation)

Item	Detail
Definition	Transforms each continuous feature to mean = 0, std = 1. Formula: $z = (x - \mu) / \sigma$. Prevents large-range features (ferritin up to 1,197) from dominating small-range ones (age 0–110).
Formula	z = (value – training_mean) / training_std
Applied to	age · d_dimer · il6 · ferritin · crp · spo2 · wbc · lymphocytes · comorbidity_count (the 9 continuous features)
Code	pipeline.py lines 277-283: scaler = StandardScaler() | X_train_scaled = scaler.fit_transform(...)
Saved as	survival_scaler.pkl — reused at inference time in app.py

3.2 Fit vs. Transform (Data Leakage Prevention)

Item	Detail
Fitting	Computing the statistics (mean, std) from data. Done ONLY on training data.
Transforming	Applying those pre-computed statistics to scale data. Applied to both train and test sets.
Why critical	Fitting on test data would let the model 'peek' at test-set statistics during training — a form of data leakage that inflates performance estimates.
Code	line 282: scaler.fit_transform(X_train...) vs. line 283: scaler.transform(X_test...)

4. Class Imbalance Terms

■ **Project Imbalance at a Glance**	96.1% Survived (8,752) vs. 3.9% Died (354) — a 24.7:1 imbalance ratio. A dummy model that always predicts 'Survived' would reach 96.1% accuracy. The meaningful metrics are Recall(Died), PR-AUC, and F1(Died).

4.1 Class Imbalance

Item	Detail
Definition	When one class has far more examples than the other. In clinical datasets this is typical because most patients survive.
This project	96.1% Survived (8,752) vs. 3.9% Died (354) — 24.7:1 ratio
Problem	High accuracy is misleadingly achievable by always predicting the majority class. Minority-class metrics are the true performance signal.

4.2 SMOTE (Synthetic Minority Over-sampling Technique)

Item	Detail
Definition	Creates synthetic (artificial, not duplicated) minority-class samples by interpolating between existing minority examples in feature space. Balances the training class distribution.
Code	pipeline.py lines 287-290: sm = SMOTE(random_state=RANDOM_STATE) | X_train_res, y_train_res = sm.fit_resample(X_train_scaled, y_train)
Applied to	Training set only — the test set is never oversampled (that would be data leakage).

5. Model Terms

5.1 Logistic Regression

Item	Detail
Definition	A linear model that estimates the log-odds of the positive class. Outputs a probability in [0, 1] via the sigmoid function. Interpretable and fast.
Strengths	Highly interpretable coefficients · fast training · well-calibrated probabilities
In project	Used as a baseline model. Achieves highest Recall(Died) = 0.9859 — misses fewest deaths.

5.2 Random Forest

Item	Detail
Definition	An ensemble of decision trees trained on random subsets of data and features. Final prediction by majority vote. Robust to outliers and non-linear relationships.
Strengths	Handles mixed feature types · resistant to overfitting · built-in feature importance
In project	Achieves Precision(Died) = 1.00 and Recall(Survived) = 1.00 — zero false alarms and no survivors missed.

5.3 XGBoost (Extreme Gradient Boosting)

Item	Detail
Definition	Gradient-boosted decision trees that iteratively correct the errors of previous trees. State-of-the-art for tabular data.
Strengths	Handles missing data natively · regularisation built-in · very high performance on structured data
Best model	Selected as best model: AUC-ROC = 0.9980 ★ | PR-AUC = 0.9999 | F1(Died) = 0.9784

5.4 k-Fold Cross-Validation

Item	Detail
Definition	Partitions training data into k equal folds; trains on k-1 folds and validates on the remaining fold, rotating k times. Gives a reliable estimate of generalisation performance.
Usage	Used during hyperparameter tuning to avoid overfitting to a single train/validation split.

6. Classification Threshold Terms

6.1 Decision Threshold

Item	Detail
Definition	The cut-point applied to a predicted probability to assign a binary class label. Default = 0.50, but can be tuned.
Example	Threshold = 0.30 → predict 'Died' for any patient with probability ≥ 30%. Increases Recall(Died) at the cost of more false positives.
Clinical relevance	In survival prediction, a lower threshold is preferred to avoid missing deaths (false negatives are more costly than false alarms).

6.2 Threshold Tuning

Item	Detail
Definition	Systematically testing many threshold values and selecting the one that optimises a chosen metric (e.g., F1(Died), Youden's J).
Trade-off	Lower threshold → higher Recall, lower Precision. Higher threshold → higher Precision, lower Recall.

7. Confusion Matrix Terms

The confusion matrix cross-tabulates actual vs. predicted labels, yielding four fundamental counts from which all classification metrics are derived.

	Predicted: Survived	Predicted: Died
Actual: Survived	TN = 68 (True Negatives)	FP = 3 (False Positives)
Actual: Died	FN = 0 (False Negatives)	TP = 1,751 (True Positives)

XGBoost confusion matrix on the 1,822-patient test set.

Item	Detail
TP (True Positive)	Model correctly predicts Survived. XGBoost: 1,751
TN (True Negative)	Model correctly predicts Died. XGBoost: 68
FP (False Positive)	Model predicts Survived but patient actually Died (false alarm). XGBoost: 3
FN (False Negative)	Model predicts Died but patient actually Survived (missed death — most dangerous error). XGBoost: 0

8. Core Classification Metrics

Item	Detail
Accuracy	Fraction of all predictions that are correct. (TP+TN) / (TP+TN+FP+FN). Misleading when classes are imbalanced.
Precision	Of all patients the model labelled Died, what fraction actually died? TP / (TP+FP). High precision = few false alarms.
Recall (Sensitivity)	Of all patients who actually died, what fraction did the model catch? TP / (TP+FN). High recall = few missed deaths.
F1-Score	Harmonic mean of Precision and Recall. 2 × (P × R) / (P + R). Balances both metrics; preferred for imbalanced classes.
Specificity	Of all patients who survived, what fraction did the model correctly label as Survived? TN / (TN+FP).

■ **Clinical Priority**

For a survival prediction system, Recall(Died) is the most safety-critical metric — it quantifies how many true deaths the model catches. A missed death (FN) is far more costly than a false alarm (FP).

9. Aggregate / Averaging Metrics

Item	Detail
Macro Average	Computes the metric for each class independently, then takes the unweighted mean. Treats all classes equally regardless of size. Useful for detecting poor performance on the minority class.
Weighted Average	Computes the metric for each class and takes a weighted mean by class support (number of samples). Dominated by the majority class. Matches overall accuracy behaviour.
Micro Average	Aggregates all TP/FP/FN counts across classes before computing the metric. Equivalent to accuracy for binary classification.

10. Ranking / Discrimination Metrics

10.1 AUC-ROC (Area Under the ROC Curve)

Item	Detail
Definition	The area under the Receiver Operating Characteristic curve. Plots True Positive Rate vs. False Positive Rate across all thresholds. AUC = 1.0 → perfect discrimination. AUC = 0.5 → random.
Best value	XGBoost: 0.9980 ★ (selected as best model on this criterion)
Interpretation	At any randomly chosen threshold, the model ranks a true 'Died' patient higher than a true 'Survived' patient with probability 0.9980.

10.2 PR-AUC (Precision-Recall Area Under Curve)

Item	Detail
Definition	The area under the Precision-Recall curve. Preferred over AUC-ROC for heavily imbalanced datasets because it focuses on minority class performance.
Best value	XGBoost & Logistic Regression: 0.9999 (tied best)
Why preferred	When positive cases are rare (3.9%), AUC-ROC can appear inflated. PR-AUC reflects the true cost of false positives among predicted positives.

11. Calibration Terms

Item	Detail
Calibration Curve	A plot of mean predicted probability (x-axis) vs. actual fraction of positives (y-axis) within probability bins. A perfectly calibrated model lies on the diagonal. Tells clinicians whether a '70% death risk' score actually corresponds to 7 out of 10 deaths.
Brier Score	Mean squared error between predicted probabilities and actual binary labels. Range 0 (perfect) to 1 (worst). For a well-calibrated model: Brier score ≈ prevalence × (1 – prevalence).
Reliability	A model is reliable if its predicted probabilities reflect true event frequencies across the full probability range. Reliability is distinct from discrimination (AUC-ROC).

12. Feature Importance & Explainability Terms

Item	Detail
SHAP Values	SHapley Additive exPlanations. Assigns each feature a contribution value for a specific prediction using cooperative game theory. Provides local (per-patient) and global (overall) explanations. The gold standard for ML interpretability.
SHAP Summary Plot	Beeswarm plot showing each feature's SHAP value for every patient. High positive SHAP → feature pushed prediction toward 'Died'. Used in app.py for the prediction explanation panel.
Permutation Importance	Measures how much model performance drops when a feature's values are randomly shuffled. Model-agnostic. Captures the real predictive contribution of a feature on held-out data.
Gini Importance	Tree-based feature importance measuring how much each feature reduces impurity (Gini or entropy) across all splits in the forest. Fast but biased toward high-cardinality and continuous features.

13. Logistic Regression Specific Terms

Item	Detail
Coefficients	The learned weights (β) for each feature. A positive coefficient increases the log-odds of the positive class; negative decreases it.
Log-Odds (Logit)	The natural logarithm of the odds ratio: log(p / (1–p)). Logistic regression models log-odds as a linear function of features.
Odds Ratio	exp(β) for each coefficient. An odds ratio of 2.0 means the feature doubles the odds of the outcome.
Regularisation (C)	Controls model complexity. $C = 1/\lambda$ where λ is the regularisation strength. Small C → strong regularisation (simpler model, less overfitting). Default C=1.0.
Sigmoid Function	$\sigma(z) = 1 / (1 + e^{-z})$. Converts log-odds to a probability in [0, 1].

14. Clinical / Domain-Specific Terms

Item	Detail
SpO2 (Oxygen Saturation)	Peripheral blood oxygen saturation measured by pulse oximetry. Normal ≥ 95%. SpO2 < 94% = hypoxia (WHO definition). Feature: low_spo2.
CRP (C-Reactive Protein)	Acute-phase protein synthesised by the liver in response to inflammation. Normal < 10 mg/L. CRP > 10 = active inflammation. Feature: high_crp.
D-Dimer	Fibrin degradation product elevated in thrombosis. Normal < 500 ng/mL (FEU). Elevated in COVID-19 coagulopathy. Feature: high_d_dimer.
Ferritin	Iron-storage protein and acute-phase reactant. Markedly elevated in severe COVID-19. Normal < 300 ng/mL. Median in cohort: 423.3 (imputed).
WBC (White Blood Cell Count)	Total white blood cell count. Indicator of immune activation. Normal: 4–11 $\times 10^3/\mu L$.
Lymphocytes	Immune cells; reduced count (lymphopenia) is a hallmark of severe COVID-19. Normal: 1.0–4.8 $\times 10^3/\mu L$. Feature: low_lymphocytes (< 1.0).
IL-6 (Interleukin-6)	Pro-inflammatory cytokine massively elevated in cytokine storm. Normal < 7 pg/mL. No IL-6 data found in the Synthea dataset → imputed with 0.
ICU (is_icu)	Indicator that the patient was admitted to an Intensive Care Unit. Strong predictor of mortality. Prevalence: 4.1% of cohort.
Comorbidities	Pre-existing conditions present before COVID-19 diagnosis. Included: hypertension (24.8%) · diabetes (31.6%) · obesity (38.1%) · COPD (1.5%) · asthma (1.9%) · heart disease (6.5%).
Survival Label	Binary target: survived = 1 if no death within 60 days of COVID diagnosis. survived = 0 if death within 60 days. 60-day window captures in-hospital + short-term post-discharge mortality.

15. Descriptive Statistics Terms

Key descriptive statistics from run_log.txt Step 4 (all 9,106 patients after imputation).

15.1 Feature Means

Feature	Mean	Interpretation
age	41.35	Average patient age = 41 years
gender	0.476	47.6% are Male (1=Male)
has_hypertension	0.248	24.8% have hypertension
has_diabetes	0.316	31.6% have diabetes
crp	10.27	Average CRP at/above inflammation threshold
was_hospitalized	0.212	21.2% were hospitalised
is_icu	0.041	4.1% admitted to ICU

15.2 Percentile Summary (Selected Features)

Feature	Q1 (25%)	Median (50%)	Q3 (75%)	Max
age	21.9	41.0	58.8	110.1
comorbidity_count	0	1	2	5
ferritin	423.3	423.3	423.3	1,197.6
d_dimer	0.3	0.3	0.3	—
wbc	—	—	—	10.5

15.3 Imbalance Ratio

Item	Detail
Definition	Ratio of majority class size to minority class size. Quantifies how skewed the class distribution is.
Formula	ratio = count(survived) / count(died)
Actual value	24.72:1 — for every 1 death, there are ~25 survivors. (run_log.txt line 20)

16. High-Level Model Comparison Summary

All key metrics from output.txt consolidated. XGBoost was selected as the best model based on highest AUC-ROC (0.9980).

Metric	Logistic Regression	Random Forest	XGBoost ★
Accuracy	0.9940	0.9984	**0.9984**
AUC-ROC	0.9975	0.9919	**0.9980 ★**
PR-AUC	0.9999	0.9993	**0.9999**
F1 Macro Avg	0.9620	0.9888	**0.9888**
F1 Weighted	0.9941	0.9983	**0.9983**
Precision (Died)	0.8750	1.0000	**1.0000**
Recall (Died)	0.9859	0.9577	**0.9577**
F1 (Died)	0.9272	0.9784	**0.9784**
Precision (Survived)	0.9994	0.9983	**0.9983**
Recall (Survived)	0.9943	1.0000	**1.0000**
F1 (Survived)	0.9969	0.9991	**0.9991**
TP	1,741	1,751	**1,751**
TN	70	68	**68**
FP	1	3	**3**
FN	10	0	**0**

■ Key Clinical Insight

Logistic Regression achieves the highest Recall(Died) = 0.9859 — it misses fewest deaths. Random Forest and XGBoost achieve Precision(Died) = 1.00 and Recall(Survived) = 1.00 — zero false alarms and no survivors missed. The optimal model choice depends on clinical priority: minimising missed deaths vs. avoiding unnecessary ICU admissions.

Quick Reference: Metric Cheat Sheet

Clinical Goal	Best Metric	Reason
Overall correctness	Accuracy	Simple but biased toward majority class
Not missing deaths	Recall (Died)	Minimises False Negatives (missed deaths)
No false death alarms	Precision (Died)	Minimises False Positives (unnecessary ICU)
Balance both	F1 (Died)	Harmonic mean of Precision + Recall
Ranking ability	AUC-ROC	Threshold-independent discrimination
Rare event focus	PR-AUC	Better than AUC-ROC for rare outcomes
Probability trustworthiness	Calibration Curve	Are probabilities what they claim?
Explainability	SHAP Values	Which features drove each prediction?

Source: pipeline.py · app.py · output.txt · run_log.txt · clinical_interpretation.txt | Dataset: Synthea 10k COVID-19 synthetic dataset | 9,106 patients | 23 features | Best model: XGBoost | AUC-ROC: 0.9980 | PR-AUC: 0.9999